

PolypNet: An End-to-End Six-Stage Deep Learning Pipeline for Colorectal Polyp Detection, Segmentation, and Malignancy Classification

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Abstract—Colorectal cancer accounts for approximately 1.9 million new cases annually worldwide, with adenoma miss rates of 17–28% during routine colonoscopy. This paper presents PolypNet, an end-to-end six-stage deep learning pipeline for automated colorectal polyp detection, segmentation, and malignancy classification from colonoscopy images. The pipeline applies image validation via a custom convolutional neural network, contrast enhancement through CLAHE, polyp localization using YOLOv11, benign/malignant classification via a three-model CNN ensemble, pixel-level segmentation using Attention U-Net with YOLO-gated masking, and structured clinical report generation using LLaVA-1.5. Trained on 74,880 annotated colonoscopy images from four public benchmarks, YOLOv11 achieves mAP@0.50 of 84.82% (95% CI: 83.1–86.5%) with 93.77% precision on 4,719 test images. The Attention U-Net, retrained with combined BCE+Dice loss, achieves a Dice coefficient of 73.41% and IoU of 64.08%, representing improvements of 12.46 and 20.25 percentage points over the BCE-only baseline. The three-model classification ensemble achieves 82.10% accuracy with 70.57% malignant recall, outperforming the best individual constituent by 0.21 percentage points. Ablation studies confirm that YOLO-gated masking contributes a 4.8 percentage point Dice improvement over unmasked segmentation, and CLAHE enhancement improves detection mAP by 1.9 percentage points over the unenhanced baseline. PolypNet is the only system in the compared literature to provide the complete workflow including input validation, detection, segmentation, classification, and natural language clinical reporting within a single deployable web application.

Index Terms—colorectal cancer, polyp detection, YOLOv11, Attention U-Net, deep learning, colonoscopy, vision language model, medical imaging, ablation study

I. INTRODUCTION

Colorectal cancer ranks as the third most frequently diagnosed malignancy in men and the second in women worldwide, accounting for approximately 1.9 million new cases and 900,000 deaths annually [1]. The adenoma-carcinoma sequence provides a clinically significant window for intervention: polyps detected and resected before malignant transformation yield five-year survival rates exceeding 90%, compared with approximately 14% for metastatic disease [2]. Colonoscopy remains the gold standard for colorectal cancer screening, yet meta-analyses consistently demonstrate that experienced gastroenterologists miss between 17 and 28% of adenomas during routine examination [3]. The miss rate for flat and sub-centimetre lesions can exceed 40%.

Artificial intelligence-assisted colonoscopy offers a compelling solution. Advances in convolutional neural networks, real-time object detectors, attention-based segmentation architectures, and vision language models have collectively made it feasible to assemble comprehensive analytical pipelines capable of supporting the clinician across the full diagnostic workflow. However, the majority of published systems address only one or two analytical tasks in isolation and present results as offline research prototypes without deployable user interfaces.

This paper presents PolypNet, an end-to-end six-stage pipeline designed to address the complete colonoscopy image interpretation workflow. The primary contributions are: (1) a robust input validation stage combining a trained CNN with domain-specific heuristics that prevents false positive findings on non-medical inputs, a mechanism largely absent from published systems; (2) a YOLO-gated segmentation strategy that constrains Attention U-Net predictions to detected bounding box regions, validated through ablation; (3) YOLOv11 applied to large-scale polyp detection with competitive mAP on a 4,719-image test set; (4) a combined BCE+Dice loss that substantially improves segmentation recall; (5) LLaVA-1.5 integration for automated natural language clinical report generation; and (6) all components deployed within a Flask web application accessible from any standard browser without programming expertise.

II. RELATED WORK

A. Polyp Detection

Early computer-aided detection systems relied on hand-crafted features and shallow classifiers [4]. Tajbakhsh et al. [5] demonstrated that CNNs trained with transfer learning from ImageNet substantially outperformed hand-crafted methods across multiple benchmarks. The YOLO architecture family [6] established single-stage real-time object detection as the dominant paradigm for colonoscopy applications. Ahmad et al. [7] benchmarked YOLOv8 on Kvasir-SEG and CVC-ClinicDB, reporting mAP@0.50 of 81.2%. Zhao et al. [8] combined YOLOv5 with a U-Net segmentation branch, achieving a detection mAP of 78.4% on CVC-ClinicDB.

B. Polyp Segmentation

Ronneberger et al. [9] introduced U-Net, establishing the encoder-decoder architecture with skip connections as the standard for biomedical image segmentation. Oktay et al. [10] extended U-Net with soft attention gates, improving segmentation of small lesions by suppressing background feature activations. Fan et al. [11] proposed PraNet with parallel partial decoders and reverse attention, achieving Dice of 89.8% on Kvasir-SEG. Transformer-based architectures such as TransFuse [12] have further advanced performance through global self-attention mechanisms.

C. Classification and Clinical Reporting

Zhang et al. [13] demonstrated that CNN ensembles produce more robust malignancy probability estimates than single models, reporting an AUC of 0.93. Template-based clinical report generation was established for radiology by Jing et al. [14]. LLaVA-1.5 [15] and BLIP-2 [16] extend this to flexible multimodal generation applicable to endoscopy reporting.

D. Research Gap

Three gaps in the existing literature motivate PolypNet. First, no published system provides the complete analytical workflow from input validation through clinical report generation in a single deployable pipeline. Second, input robustness through dedicated validation mechanisms is largely absent from published prototypes. Third, accessible web interfaces for clinical operators remain rare in the polyp AI literature.

III. SYSTEM ARCHITECTURE

PolypNet processes each colonoscopy image through six sequential stages. The pipeline terminates early if any stage determines that further processing is unwarranted, conserving computation and preventing misleading outputs.

A. Stage 1: Image Validation

The validation module combines a custom CNN classifier with seven deterministic heuristic checks. The CNN is a four-block architecture with progressive channel counts of 32, 64, 128, and 256, followed by global average pooling and two fully connected layers. It was trained on a balanced dataset of 7,400 negative images — comprising CIFAR-10 photographs, synthetic brain MRI patterns, and facial images — alongside 7,400 positive colonoscopy frames, achieving 100% accuracy on the held-out validation set across 25 training epochs.

The seven heuristic checks enforce domain-specific constraints: red channel dominance consistent with mucosal tissue colouration, characteristic dark vignette border detection, colour spread thresholds, near-grayscale rejection, blue channel dominance rejection, brightness ceiling, and high saturation rejection. Either the CNN prediction or any heuristic check failing causes immediate pipeline termination with an informative rejection message.

B. Stage 2: CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied in the LAB colour space on the luminance channel only, preserving natural tissue chrominance. The technique subdivides the image into contextual tiles and performs local histogram equalization with a clip limit parameter that prevents noise amplification in low-variance regions. Bilinear interpolation between tile boundaries ensures smooth spatial transitions. Ablation results in Section V-D quantify the detection improvement attributable to this stage.

C. Stage 3: YOLOv11 Detection

The enhanced image is processed by a fine-tuned YOLOv11-nano model [24] comprising 2.58 million parameters and 6.3 GFLOPs at 640×640 pixel input resolution. YOLOv11 introduces C3k2 cross-stage partial network blocks in the backbone that improve gradient flow and feature reuse, and an anchor-free detection head well suited to irregular polyp morphologies. The path aggregation neck provides multi-scale context for both large sessile lesions and sub-centimetre flat polyps.

The model was initialised with COCO pre-trained weights and fine-tuned for 150 epochs using cosine annealing from 0.01 to 10^{-4} . Training incorporated mosaic augmentation, random flipping, and colour jitter. Detections exceeding a confidence threshold of 0.35 are passed to Stages 4 and 5. If no detection exceeds this threshold, the pipeline bypasses classification and segmentation and proceeds directly to a no-polyp clinical report.

D. Stage 4: CNN Classification Ensemble

The classification stage employs a three-model ensemble comprising VGG16 [21], ResNet50 [22], and EfficientNetB0 [23]. Each model was initialised with ImageNet pre-trained weights and fine-tuned using Adam at 10^{-4} with binary cross-entropy loss and early stopping at patience 15. The ensemble prediction is the arithmetic mean of the three models' malignant class probabilities, with a decision threshold of 0.50. Individual model performance versus ensemble performance is reported in Section V-D.

E. Stage 5: Attention U-Net Segmentation

The Attention U-Net [10] generates a pixel-level probability map of polyp regions through an encoder-decoder with attention-gated skip connections. Each attention gate computes a soft coefficient per spatial location, amplifying polyp-relevant activations while suppressing background tissue.

The model was trained with a combined loss:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{BCE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}} \quad (1)$$

where the Dice loss is:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + 1}{\sum_i p_i + \sum_i g_i + 1} \quad (2)$$

with p_i and g_i denoting predicted and ground truth pixel values respectively.

A YOLO-gated masking procedure zeros all predicted pixels outside the union of Stage 3 bounding boxes before thresholding, preventing spurious whole-image masks. Post-processing applies morphological opening with a 5×5 elliptical kernel, largest connected component filtering, and morphological closing with a 3×3 kernel. The contribution of YOLO-gated masking is quantified in Section V-D.

F. Stage 6: VLM Clinical Report Generation

The final stage generates a structured clinical report by passing the colonoscopy image with segmentation overlay to LLaVA-1.5-7B [15], combining a CLIP visual encoder with the Vicuna language backbone. The clinical prompt uses the USER/ASSISTANT conversation format and requests assessment of polyp size, morphology, surface characteristics, and recommended follow-up.

A three-tier fallback chain is implemented: LLaVA-1.5 (primary, requires 8 GB VRAM), BLIP-2 [16] (secondary), and a structured template (always available). A safety guard prevents polyp-related clinical language from being generated when Stage 3 returns no detections.

IV. EXPERIMENTAL SETUP

A. Dataset

The training corpus comprises 74,880 annotated colonoscopy images assembled from four publicly available benchmarks. Table I provides the exact composition. Kvasir-SEG [17] provides pixel-level segmentation masks annotated by experienced gastroenterologists from Vestre Viken Health Trust, Norway. CVC-ClinicDB [18] provides ground truth masks derived from 31 colonoscopy sequences. ETIS-LaribPolypDB [19] contributes 196 images specifically curated to challenge detection of flat and sub-centimetre polyps. The broader Kvasir dataset [20] contributes classification-labelled images covering benign and malignant polyp categories.

TABLE I
DATASET COMPOSITION AND EXACT SPLIT COUNTS

Source	Total	Train	Val	Test
Kvasir-SEG [17]	1,000	700	150	150
CVC-ClinicDB [18]	612	428	92	92
ETIS-LaribPolypDB [19]	196	137	30	29
Kvasir (extended) [20]	73,072	51,150	10,961	10,961
Total	74,880	52,415	11,233	11,232
Detection test set (YOLO)			4,719 images	

The dataset was partitioned using stratified sampling (70% train / 15% val / 15% test). The test subset was held out entirely during model development and used exclusively for final reported evaluation. All images were resized to model-specific input dimensions and normalised using ImageNet mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225]. Training augmentation included random horizontal and vertical flipping, rotation within $\pm 15^\circ$, brightness and contrast jitter within $\pm 20\%$, and Gaussian blur. YOLOv11 additionally used mosaic augmentation.

B. Training Configuration

All training was conducted on an NVIDIA RTX 4090 GPU (24 GB VRAM). Table II summarises hyperparameters.

TABLE II
TRAINING CONFIGURATION FOR ALL POLYPNET COMPONENTS

Model	Ep.	BS	LR	Loss	Time
YOLOv11	150	16	0.01	CIoU+BCE	~2h
Attn U-Net	150*	8	1e-4	BCE+Dice	~8h
VGG16	17**	32	1e-4	BCE	~1h
ResNet50	17**	32	1e-4	BCE	~1.5h
EffNetB0	17**	32	1e-4	BCE	~1h
Val. CNN	25	32	1e-3	BCE	~20m

*Early stopping, patience 15

**Early stopping triggered at epoch 17 of 50

C. Evaluation Metrics

Detection performance is reported using mAP@0.50 and mAP@0.50:0.95, precision, recall, and F1 score. Segmentation is reported using Dice and IoU at the optimal threshold determined by sweep from 0.50 to 0.70 in increments of 0.01. Classification is reported using overall accuracy, per-class recall, precision, and F1. Confidence intervals (95%) for primary metrics are computed using bootstrap resampling with 1,000 iterations on the held-out test set.

V. RESULTS AND DISCUSSION

A. Detection Results

Table III presents YOLOv11 detection results on the 4,719-image held-out test set with 95% confidence intervals.

TABLE III
YOLOV11 DETECTION RESULTS — 4,719 TEST IMAGES

Metric	Value	95% CI
mAP@0.50	84.82%	[83.1, 86.5]
mAP@0.50:0.95	60.32%	[58.4, 62.2]
Precision	93.77%	[92.4, 95.1]
Recall	73.00%	[71.2, 74.8]
F1 Score	82.09%	[80.6, 83.6]
Inference Speed	0.4 ms/image	—

The precision of 93.77% is the most clinically significant result, indicating that fewer than 7% of generated bounding boxes correspond to false positive detections. In a clinical context, false positive detections could trigger unnecessary biopsy procedures. The recall of 73.00% reflects the inherent difficulty of detecting small and flat polyp morphologies under white-light imaging. The F1-confidence curve peaks at 0.88 at a confidence threshold of 0.469, confirming stable performance across a wide operating range.

B. Segmentation Results

Table IV compares Attention U-Net performance under BCE-only loss against the retrained BCE+Dice formulation.

The combined BCE+Dice loss improved Dice by 12.46 and IoU by 20.25 percentage points. The recall gain of 24.95 percentage points demonstrates that the Dice loss term provided

TABLE IV
ATTENTION U-NET SEGMENTATION RESULTS

Metric	BCE Only	BCE+Dice (Ours)
Dice Score	60.95%	73.41%
IoU	43.83%	64.08%
Precision	86.88%	84.13%
Recall	46.94%	71.89%
Optimal Threshold	0.52	0.50

the direct gradient signal for foreground pixel sensitivity that pure BCE training could not deliver under class imbalance between background and polyp pixels.

C. Classification Results

Table V presents three-model ensemble results alongside individual model performance.

TABLE V
CLASSIFICATION RESULTS: INDIVIDUAL MODELS VS ENSEMBLE

Model	Acc.	Ben.R	Mal.R	Mal.F1
VGG16	81.89%	88.40%	68.58%	74.21%
ResNet50	69.70%	77.19%	43.39%	54.89%
EfficientNetB0	70.44%	74.19%	65.84%	65.43%
Ensemble	82.10%	90.61%	70.57%	77.01%

The ensemble achieves the highest overall accuracy (82.10%), benign recall (90.61%), malignant recall (70.57%), and malignant F1 (77.01%) across all configurations, confirming that combining architectures with diverse inductive biases produces more robust predictions than any single constituent. ResNet50 shows the weakest malignant recall at 43.39%, highlighting the benefit of ensemble averaging in reducing individual model blind spots. Malignant recall is the most clinically critical metric, as failure to identify a malignant lesion carries greater consequence than a false positive referral.

D. Ablation Study

Table VI reports the contribution of CLAHE enhancement and YOLO-gated masking through controlled component removal experiments on the held-out test set.

TABLE VI
ABLATION STUDY: COMPONENT CONTRIBUTIONS

Configuration	Det. mAP@50	Seg. Dice
Full PolypNet	84.82%	73.41%
w/o CLAHE (raw image)	82.93%	72.18%
w/o YOLO-gated mask	84.82%	68.61%
w/o CLAHE + w/o mask	82.93%	64.30%

Removing CLAHE reduces detection mAP by 1.89 percentage points, confirming that luminance-only contrast enhancement provides a meaningful preprocessing benefit for the downstream detector without distorting tissue chrominance. Removing YOLO-gated masking reduces segmentation Dice

by 4.80 percentage points, demonstrating that constraining U-Net predictions to detected bounding box regions substantially improves mask quality by preventing whole-image false positive masks. The combined removal of both components degrades Dice to 64.30%, a reduction of 9.11 percentage points from the full system, confirming that the two components provide complementary and partially independent contributions.

E. Comparison with Related Works

Table VII compares PolypNet against published systems. Direct numerical comparison must be interpreted with reference to dataset and protocol differences across studies.

TABLE VII
COMPARISON WITH PUBLISHED BASELINE SYSTEMS

Method	mAP@50	Dice	E2E	VLM
Tajbakhsh et al. [5]	88.3% [†]	–	No	No
Fan et al. PraNet [11]	–	89.8%	No	No
Zhao et al. [8]	78.4%	71.2%	Partial	No
Ahmad et al. [7]	81.2%	–	No	No
PolypNet (ours)	84.82%	73.41%	Yes	Yes

[†]Reported as sensitivity, MICCAI 2015 dataset

PolypNet achieves a higher detection mAP than Ahmad et al. and Zhao et al. on their respective datasets. PraNet achieves a higher Dice of 89.8%; however, it is a dedicated segmentation-only model without detection, classification, or reporting capability, and is evaluated on a smaller dataset under different protocols. PolypNet is the only system in this comparison providing the complete end-to-end analytical workflow with VLM clinical report generation and a deployable web interface. These comparisons should be treated as indicative rather than definitive given differences in dataset composition and evaluation protocols.

F. System-Level Evaluation

Manual testing on 100 diverse cases (60 polyp colonoscopy images, 20 non-medical images, 20 edge cases including gastric endoscopy and medical illustrations) yielded 96 correct outcomes. The validation CNN correctly rejected 18 of 20 non-medical images; two edge cases passed due to atypical red-channel profiles. LLaVA-1.5-7B successfully generated coherent clinical language in all 30 VLM test runs, with the safety guard correctly preventing polyp-related language in all no-detection cases.

VI. CONCLUSION

This paper presented PolypNet, a six-stage end-to-end deep learning pipeline for automated colorectal polyp detection, segmentation, and malignancy classification. Ablation studies confirm the contribution of CLAHE enhancement (+1.89% mAP) and YOLO-gated masking (+4.80% Dice) to overall system performance. The combined BCE+Dice loss improves segmentation Dice from 60.95% to 73.41%. The three-model ensemble outperforms all individual constituents on malignant recall, the most clinically relevant classification metric.

YOLOv11 achieves 84.82% mAP@0.50 with 93.77% precision on 4,719 test images (95% CI: 83.1–86.5%). The Attention U-Net achieves 73.41% Dice and 64.08% IoU. The three-model ensemble achieves 82.10% accuracy with 70.57% malignant recall. All components are integrated within a deployable Flask web application.

Limitations include the absence of prospective clinical validation, the relatively lower malignant recall of 70.57% compared to clinical requirements for screening tools, and evaluation on benchmark datasets that may not fully represent the variability of real-world endoscopy equipment and patient populations.

Future work will target real-time video stream processing, PACS/DICOM integration, multi-class polyp type classification, prospective clinical validation, and cross-centre generalisation studies.

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